DIVYANSHU SINGH CHAUHAN

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Robotics & Autonomous Systems Researcher | Python, C++, ROS2, Optimal Control

Education

University of Michigan, Ann Arbor — Michigan, USA
MSE - Aerospace Engineering (GPA: 3.7/4.0); (Subdomain: Autonomous Systems and Control) — *Aug 2024 - May 2026*

Indian Institute of Technology (IIT) Guwahati — Guwahati, India
B.Tech - Mechanical Engineering (GPA: 8.27/10.0) — *July 2017 - July 2021*

Relevant Coursework: Optimal & Nonlinear Control, Trajectory Optimization, State Estimation, Robotics Systems, Learning-Based Control

Experience

Python Developer, Standard Chartered Bank — Aug 2021 - Jul 2024

- Implemented and deployed Python-based SOAR automation using **Splunk Phantom**, reducing incident response time by **40%** and analyst workload by **30%** across production security pipelines.
- Built **RESTful** microservices using **FastAPI**, integrated with enterprise security tooling; maintained production codebases using **Git** and **Jira-driven Agile workflows**.
- Co-developed a **supervised phishing detection model (83% accuracy)**, contributing feature engineering and model evaluation pipelines.

Deep Learning Summer Intern, HyperVerge Inc. — May 2020 - Aug 2020

- Developed **U-Net** and **GAN-assisted autoencoders** for image tampering localization, achieving **91%** detection accuracy on authentic data and significantly improving localization performance on manipulated samples.
- Designed a custom **Conv2D** layer that improved image manipulation detection, increasing precision rates from **75%** to **90%**, and allowing for more accurate insights in data-driven decision-making processes.

Projects

Robust Trajectory Optimization in Orbital Mechanics under Uncertainty — May 2025 - Ongoing
Prof. Alex Gorodetsky & CFD Research

- Developed an end to end **three phase** trajectory pipeline for autonomous docking spacecraft on a **deterministic** target location under the actual constraints as per NASA's Docking standards using **CasADI** framework and **IPOPT** numerical solver.
- Formulating terminal constraints using uncertainty-aware cost functions, incorporating statistical bounds via **Monte Carlo sampling and covariance-based quaternion perturbations**.
- Designed the trajectory framework to support belief-space planning and future integration of EKF/UKF-based onboard state estimation.
- **Skills**: Python, NumPy, CasADi, State Estimation, Trajectory Optimization

Communication-Aware, Physics-Based and Photo-Realistic simulator for multi-robot systems — Jan 2025 - May 2025
Prof. Vasileios Tzoumas, University of Michigan

- Developed multi-robot simulations for semantic mapping using **ROS2** for middleware in **C++** language in **Unity** Environment.
- Worked on migration of the project from **ROS1** to **ROS2** with the implementation of **SlideSLAM**.
- Explored active spatial perception and map reconstruction methods within a multi-robot simulation framework.

Botlab and Armlab (BotLab & ArmLab) — Aug 2024 - Nov 2024

- Implemented **A* planning** and **particle-filter SLAM** on mBot using **IMU** and **LiDAR** sensors.
- Developed forward and inverse kinematics for a **5-DOF** robotic arm; integrated **RealSense-based vision control** in Python.

Lunar Anomaly Search — May 2019 - Aug 2019

- Developed an end to end **keras/tensorflow** pipeline with a team of researchers to find anomalies on the lunar surface using **variational autoencoder models**.
- Utilized **terabytes** of real-time data from NASA's LRO camera to train the deep learning model on google cloud.

Technical Skills

- **Languages**: Python, C++, MATLAB
- **Robotics Control Tools**: ROS1, ROS2, Simulink, CasADi, PID control, SLAM, Kalman Filter (EKF, UKF), Particle Filter
- **Planning & Estimation Algorithms:** A*, MPC, Probabilistic Inference, Trajectory Optimization
- **Machine Learning:** TensorFlow, Keras, GANs, Autoencoders, KNN, SVM, Computer Vision
- **Other Tools**: Git, Linux, FastAPI, Splunk Phantom

Publications

Deep Learning via LSTM Models for COVID-19 Infection Forecasting in India, PLOS One Link

Neural Anomaly Search on Lunar Reconnaissance Orbiter Camera Images, AbSciCon 2022 Link